

Reference-Measure Geometry in Quantum Parameter Estimation: When Coordinate Surrogates Produce Ridge-Side Convergence

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We examine the role of reference-measure selection in quantum parameter estimation on curved manifolds, with emphasis on single-qubit gate estimation on $SU(2)$. In many contemporary calibration and estimation workflows, intrinsically defined likelihoods are combined with priors or regularization terms specified in local Euclidean coordinates. When isotropy or noninformativeness is intended with respect to the intrinsic (Haar/Riemannian) geometry of the group, this practice implicitly replaces the Haar reference measure with Lebesgue measure, thereby specifying a different statistical model rather than a reparametrization of the same one. We show that omitting the associated chart-volume factor alters the optimization objective itself, modifying its gradient field and stationary-point structure. This mismatch persists arbitrarily close to the identity and produces systematic stationary-point misalignment: flat-coordinate surrogate objectives can converge to points that are non-stationary for the corresponding Haar-consistent objective. We demonstrate that this effect arises in realistic gate-estimation pipelines and can affect strictly interior solutions, particularly under moderate-to-strong regularization or limited data. Euclidean-coordinate models remain appropriate when physical noise or prior structure is genuinely coordinate-Gaussian, but our results apply specifically to workflows where intrinsic isotropy is intended yet implemented using flat-coordinate surrogates. More broadly, the analysis identifies reference-measure selection as a modeling decision with direct consequences for optimization and inference on curved parameter manifolds.

I. INTRODUCTION

Modern quantum estimation and control workflows routinely combine two distinct geometric structures. Forward models evolve quantum states or gates on curved manifolds such as $SU(2)$ using intrinsically defined physical laws and likelihoods [1, Ch. 3]. Optimization, however, is typically performed in local coordinate charts $\theta = (\theta_1, \dots, \theta_d) \in \mathbb{R}^d$ using Euclidean objectives built from priors, noise models, or regularization terms specified directly in those coordinates [2, 3]. This practice implicitly adopts Lebesgue measure on $\mathbb{R}^d$ as the reference measure for inference and optimization.

When probability models are defined intrinsically with respect to Haar (bi-invariant Riemannian) volume on $SU(2)$, their evaluation in local coordinates necessarily includes a chart-volume factor given by the square root of the metric determinant. Omitting this factor does not merely correspond to a coordinate reparametrization of the same statistical model; it specifies a different model through a different choice of reference measure. As a consequence, the resulting objective differs not only in curvature but in its gradient field and stationary-point structure.

This distinction is operationally important. Flat-coordinate surrogate objectives can converge to points that satisfy standard optimizer tolerances while being

non-stationary for the corresponding Haar-consistent objective. Coordinate-gradient ascent directions can therefore correspond to descent directions under the intrinsic geometry (see Proposition 1), and coordinate-space optima can lie on ridges rather than minima of the intrinsic loss landscape.

This paper shows that such reference-measure mismatch produces systematic optimization artifacts in realistic quantum gate-estimation workflows. We demonstrate that the mismatch persists arbitrarily close to the identity, inducing a second-order correction to local curvature-based quantities such as Fisher information. More importantly, we show that it alters global optimization behavior: flat-coordinate surrogate objectives can converge to points that are not stationary for the intended Haar-consistent objective, even in regimes where local Gaussian approximations are commonly assumed to be valid.

Our contribution is methodological rather than geometric. While the differential geometry is classical, we show that reference-measure inconsistency systematically distorts optimization in modern gate-estimation pipelines, producing stationary-point misalignment, ridge-side convergence, and boundary-amplified failures under realistic regularization and data regimes.

We make this mechanism explicit in three steps. First, a global illustration using stereographic projection (Appendix A) shows how intrinsically uniform densities become highly nonuniform in coordinates solely through the chart-volume factor. Second, a local analysis in exponential (Lie-algebra) coordinates (Section III) quantifies the induced second-order correction to curvature-based diag-

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nostics and shows that the mismatch persists arbitrarily close to the identity. Third, large-scale multi-start single-qubit gate-estimation experiments (Section IV) demonstrate that flat-coordinate optima are generically non-stationary for the Haar-consistent objective in realistic optimization regimes.

Euclidean-coordinate models are appropriate when physical noise or prior structure is genuinely coordinate-Gaussian, such as additive perturbations introduced by control hardware. Our analysis concerns workflows in which intrinsic isotropy is intended but implemented using flat-coordinate surrogates. In this setting, reference-measure selection becomes a modeling decision with direct consequences for optimization and inference on curved parameter manifolds.

Figure 1 summarizes the key objects, coordinate systems, and conventions used throughout. We distinguish Euclidean (flat-coordinate) quantities with subscript E from geometric (Haar-consistent) quantities with subscript G .

II. FORWARD-MODEL GEOMETRY VERSUS OPTIMIZATION GEOMETRY

Let M be a smooth n -dimensional Riemannian manifold equipped with metric tensor g and associated volume measure $d\mu_g$. Intrinsic probability densities on M are defined with respect to this Riemannian volume form. If $\psi : V \subset \mathbb{R}^n \rightarrow M$ is a smooth coordinate chart, the pull-back of the volume measure to coordinate space satisfies

$$\psi^*(d\mu_g) = \sqrt{\det g(y)} dy, \quad (1)$$

where $g(y)$ is the matrix of metric components in the chosen coordinates and dy denotes Lebesgue measure on $\mathbb{R}^n$ (see, e.g., [12, Ch. 15, p. 390]; cf. Appendix B).

In quantum estimation and control workflows, likelihood functions are often defined intrinsically on curved manifolds such as $SU(2)$ or $SU(2^n)$, while optimization is carried out in local coordinates using Euclidean objectives and classical optimizers. Priors, noise models, and regularization terms are therefore most often specified directly in coordinate space, implicitly adopting Lebesgue measure as the reference measure for inference and optimization.

This choice is sometimes physically appropriate. For example, control electronics may introduce approximately additive Gaussian perturbations in a chosen parameterization, in which case a flat-coordinate prior accurately reflects the underlying noise structure. However, when isotropy or noninformativeness is intended with respect to the intrinsic (Haar/Riemannian) geometry of the group, specifying priors or regularizers in flat coordinates without the associated chart-volume factor encodes a different statistical model. In that case, the resulting objective is not a reparametrization of an intrinsic likelihood, but a surrogate defined with respect to a different reference measure.

The present work concerns this modeling choice: the reference measure used to define probability on parameter space. When intrinsically defined likelihoods are combined with coordinate-space regularization while intending intrinsic isotropy, omitting the chart-volume factor alters the objective itself, changing its gradient field and stationary-point structure. The following sections quantify the consequences of this reference-measure mismatch, first through local analysis in Lie-algebra coordinates and then through numerical optimization experiments in single-qubit gate estimation.

In the next section, we show how this abstract reference-measure distinction manifests concretely in exponential coordinates on $SU(2)$, where the Jacobian can be written in closed form and its optimization consequences made explicit.

III. LIE-ALGEBRA ANALYSIS IN EXPONENTIAL COORDINATES

We now make the reference-measure mismatch explicit in exponential (Lie-algebra) coordinates for $SU(2)$. This local analysis isolates the geometric source of distortion and shows that it persists arbitrarily close to the identity, where flat-coordinate approximations are often assumed to be reliable.

Exponential coordinates and Haar volume

Let $\exp : \mathfrak{su}(2) \rightarrow SU(2)$ denote the Lie-group exponential map. In the Pauli basis, any element of $SU(2)$ sufficiently close to the identity admits a unique representation as the exponential of a Lie algebra element, which can be written as $\exp(iv \cdot \sigma/2)$ with $v \in \mathbb{R}^3$ [4, Sec. 3.4] [1, p. 218]. Thus,

$$U(v) = \exp\left(\frac{i}{2} v \cdot \sigma\right), \quad v \in \mathbb{R}^3, \|v\| < 2\pi,$$

where $\sigma = (\sigma_x, \sigma_y, \sigma_z)$ are the Pauli matrices.

These exponential coordinates identify a neighborhood of the identity in $SU(2)$ with a ball in $\mathbb{R}^3$. In this chart, the bi-invariant Riemannian volume (Haar measure) pulls back to Lebesgue measure with a scalar Jacobian factor $J_{\exp}(v) = \sqrt{\det g(v)}$ given by

$$J_{\exp}(v) = \left(\frac{\sin(\|v\|/2)}{\|v\|/2}\right)^2 \quad (2)$$

so that

$$d\mu_{\text{Haar}}(U(v)) = J_{\exp}(v) dv,$$

where dv denotes Lebesgue measure on $\mathbb{R}^3$ [4, Ch. 4]; see also [13]. The apparent singularity at $v = 0$ is removable, with $J_{\exp}(0) = 1$ by continuity.

Notation Summary

Manifold, Measures

- $SU(2) \cong S^3$, group elements $\omega \in S^3$
- μ_{Haar} : Haar (bi-invariant) measure on $SU(2)$
- μ_{Leb} : Lebesgue measure on $\mathbb{R}^3$

Coordinates and Maps

- $v \in \mathbb{R}^3$: exponential (axis-angle) coordinates, $\|v\| < 2\pi$
- $y \in \mathbb{R}^3$: stereographic coordinates
- $U(v) = \exp(iv \cdot \sigma/2)$: exponential map $\mathbb{R}^3 \rightarrow SU(2)$
- $\text{Log} : SU(2) \setminus \{-I\} \rightarrow \mathbb{R}^3$: inverse of U (defined except at antipodal point)
- $\pi : S^3 \setminus \{N\} \rightarrow \mathbb{R}^3$: stereographic projection

Geometry

- g : bi-invariant Riemannian metric on $SU(2)$
- $J_{\text{exp}}(v) = \left(\frac{\sin(\|v\|/2)}{\|v\|/2} \right)^2$: exponential chart volume factor
- $J_{\pi}(y) = \left(\frac{2}{1+\|y\|^2} \right)^3$: stereographic chart volume factor
- In any chart: $d\mu_{\text{Haar}} = J_{\text{chart}} d(\text{coordinates})$

Probability Densities

- $f_E(v)$: Euclidean (flat-coordinate) density on $\mathbb{R}^3$
- $f_G(v) \propto f_E(v) J_{\text{exp}}(v)$: Haar-consistent density
- f_E and f_G are Lebesgue densities inducing different measures on $SU(2)$

Objectives

- $L_D(v)$: data loss (negative log-likelihood)
- $L_E(v)$: Euclidean objective
- $L_G(v) = L_E(v) - \log J_{\text{exp}}(v)$: Haar-consistent objective

Conventions

- ∇ : Euclidean gradient in coordinates
- $\|\cdot\|$: Euclidean norm unless stated otherwise
- Subscripts E (Euclidean/flat) vs. G (Geometric/Haar)

FIG. 1. Summary of notation and conventions used throughout this work.

Euclidean versus Haar-consistent coordinate densities

Let $f_E(v)$ denote a probability density on $\mathbb{R}^3$ defined with respect to Lebesgue measure dv , for example a zero-mean isotropic Gaussian. Viewed as a coordinate-space density, f_E induces a probability measure on $SU(2)$ via the exponential map. This induced measure is not Haar unless f_E is modified by the chart-volume factor.

The density with respect to Haar measure that corresponds to the Lebesgue-coordinate density f_E is therefore

$$f_G(v) = C f_E(v) J_{\text{exp}}(v), \quad (3)$$

where C is a normalization constant. This relation follows from the change-of-variables formula on manifolds together with the fact that the Haar measure on $SU(2)$ pulls back under the exponential map to $d\mu_{\text{Haar}}(U(v)) = J_{\text{exp}}(v) dv$ (see, e.g., [4, Ch. 4]; [12, Chs. 15–16]). Although f_E and f_G are both ordinary Lebesgue densities on $\mathbb{R}^3$, they represent densities with respect to different reference measures on $SU(2)$.

Local expansion near the identity

To quantify the mismatch locally, expand the logarithm of the Jacobian in Eq. (2) about $v = 0$. Using the Maclaurin series $\sin x = x - x^3/6 + O(x^5)$, one obtains

$$-\log J_{\text{exp}}(v) = \frac{1}{12} \|v\|^2 + O(\|v\|^4). \quad (4)$$

Thus, even arbitrarily close to the identity, omitting the chart-volume factor introduces a second-order correction to the log-density.

Combining Eq. (3) with the local expansion (4) yields

$$\log f_G(v) = -\frac{1}{2\tau^2} \|v\|^2 - \frac{1}{12} \|v\|^2 + O(\|v\|^4) + \text{const.} \quad (5)$$

Consequences for curvature diagnostics

Because the Jacobian contributes at second order, curvature-based quantities computed from flat-coordinate surrogates differ systematically from those computed from Haar-consistent models. In particular, Fisher information computed from f_E and f_G disagrees at $O(\tau^{-2})$ even near the identity, despite both models being locally Gaussian in coordinates.

This discrepancy reflects the fact that the Jacobian modifies the log-density itself rather than merely rescaling coordinates. The difference therefore appears directly in gradients and Hessians, not only in higher-order corrections.

Implications for optimization objectives

The same mechanism carries over to regularized optimization objectives. A typical Euclidean surrogate loss (negative log-likelihood with a ℓ_2 penalty) of the form

$$L_E(v) = -\log \mathbb{P}(D \mid U(v)) + \frac{\lambda}{2} \|v\|^2$$

corresponds to a Gaussian penalty defined in flat parameter space and is therefore naturally interpreted with respect to a Lebesgue reference measure.

The objective corresponding to the Haar-consistent density is

$$L_G(v) = -\log \mathbb{P}(D \mid U(v)) + \frac{\lambda}{2} \|v\|^2 - \log J_{\text{exp}}(v).$$

Differentiating yields

$$\nabla L_G(v) = \nabla L_E(v) - \nabla \log J_{\text{exp}}(v),$$

where $L_E(v) = -\log \mathbb{P}(D \mid U(v)) + \frac{\lambda}{2} \|v\|^2$ is the Euclidean objective. The additional term contributes a nonzero gradient that modifies the stationary-point equations even arbitrarily close to $v = 0$, since $\log J_{\text{exp}}(v)$ departs from zero at quadratic order in $\|v\|$ and therefore produces a linear-order correction in its gradient.

As a result, solutions that satisfy $\nabla L_E(v^*) = 0$ need not satisfy $\nabla L_G(v^*) = 0$. This local mismatch underlies the global stationary-point misalignment observed in the numerical experiments of Section IV.

IV. ANALYTICAL EVALUATION

This section makes the reference-measure issue concrete in three complementary ways. We first quantify the local Jacobian-induced correction in exponential coordinates via an analytic Laplace prediction and a score-based Fisher-information check that isolates reference-measure effects. We then show that the same Jacobian term generically changes stationary points of regularized coordinate-space objectives, implying that convergence of a flat-coordinate optimizer does not certify stationarity of the Haar-consistent objective. Finally, we quantify how often this mismatch matters operationally in a multi-start single-qubit gate-estimation study that mirrors standard practice in tomography and calibration [5, 6].

Local magnitude: Laplace prediction and score-based Fisher-information validation

The flat and geometric formulations are both functions on the same coordinate chart $v \in \mathbb{R}^3$; they differ only in the induced measure on $SU(2)$ (Lebesgue in coordinates versus Haar on the group). In exponential (Lie-algebra) coordinates, the Haar chart-volume factor is equation 2, so the geometry-consistent small-noise local density (in coordinates) can be written as

$$f_G(v) \propto \exp\left(-\frac{\|v\|^2}{2\tau^2}\right) J_{\text{exp}}(v), \quad (6)$$

with $\tau > 0$ controlling concentration around the identity.

Samples $\xi_\ell \in \mathbb{R}^3$ from (6) are generated by rejection sampling using the Euclidean Gaussian envelope

$$f_E(v) = (2\pi\tau^2)^{-3/2} \exp\left(-\frac{\|v\|^2}{2\tau^2}\right)$$

as a proposal and acceptance probability $J_{\text{exp}}(v)$. This is valid because $(\sin x)/x \leq 1$ implies $J_{\text{exp}}(v) \leq 1$ on the chart domain.

Monte Carlo estimators are formed under the coordinate-space density consistent with the objective being evaluated: Euclidean samples for L_E -based quantities and Haar-consistent samples for L_G -based quantities. This avoids conflating reference-measure mismatch with sampling-measure mismatch when comparing losses, gradients, or Fisher information.

We validate the geometric correction directly at the level of Fisher information, which probes the same local quadratic (Laplace) structure derived in Eq. (5), rather than by fitting τ from moments. For a location family $f(v - \mu)$, the score identity gives

$$\nabla_\mu \log f(v - \mu) \Big|_{\mu=0} = -\nabla_v \log f(v),$$

so the Fisher information at $\mu = 0$ is estimated by the score outer product

$$\hat{I} = \frac{1}{N} \sum_{\ell=1}^N s(\xi_\ell) s(\xi_\ell)^\top, \quad s(v) = \nabla_v \log f(v). \quad (7)$$

This targets the same local quadratic (Laplace) structure as Eq. (5) and avoids ambiguities associated with variance fitting under misspecified reference measures.

Using the same samples drawn from f_G , we evaluate two score functions corresponding to two *assumed* models: the Euclidean surrogate f_E and the geometry-consistent model $f_G \propto f_E J_{\text{exp}}$. Their scores are

$$\nabla_v \log f_E(v) = -\frac{1}{\tau^2} v,$$

$$\nabla_v \log f_G(v) = -\frac{1}{\tau^2} v + \nabla_v \log J_{\text{exp}}(v).$$

TABLE I. Score-based Fisher-information validation under geometry-consistent sampling. All samples are drawn from the geometry-consistent model $f_G(v)$ ($N = 2 \times 10^4$ per τ). Fisher information for the location parameter μ at $\mu = 0$ is estimated by $\hat{I} = \frac{1}{N} \sum_{\ell=1}^N s(\xi_\ell) s(\xi_\ell)^\top$, where $s(v) = \nabla_v \log f(v)$ is the score function. The two columns differ only in which score is used: $\hat{I}_E$ uses the Euclidean score $s_E(v) = -v/\tau^2$ (derived from f_E but evaluated on samples from f_G), while $\hat{I}_G$ uses the geometric score $s_G(v) = -v/\tau^2 + \nabla_v \log J_{\text{exp}}(v)$ (the correct score for f_G). The analytical prediction is $I_G \approx 1/\tau^2 + 1/6$ and $I_E = 1/\tau^2$, giving the predicted ratio $(I_G/I_E)_{\text{pred}} = 1 + \tau^2/6$. This leading-order small-noise (Laplace) diagnostic is validated for $\tau \lesssim 0.2$; systematic deviations at larger τ reflect higher-order geometric effects beyond the quadratic approximation. Because $\hat{I}_E$ is computed using the Euclidean score on samples from f_G , it quantifies the model misspecification error from omitting the Jacobian, not the Fisher information of the Euclidean model under Euclidean sampling.

All samples are drawn from $f_G(v)$ with $N = 2 \times 10^4$.				
τ	Eucl. score $\hat{I}_E$	Geom. score $\hat{I}_G$	Ratio $\hat{I}_G/\hat{I}_E$	Predicted $(I_G/I_E)_{\text{pred}}$
0.05	399.50	399.83	1.0008	1.0004
0.10	100.73	101.06	1.0033	1.0017
0.20	24.73	25.06	1.0134	1.0067
0.50	3.82	4.16	1.0869	1.0417
1.00	0.85	1.19	1.3934	1.1667

We denote the resulting Monte Carlo Fisher estimates by $\hat{I}_E$ and $\hat{I}_G$ (obtained by inserting the corresponding score into (7)). Throughout, τ is treated as fixed and known when evaluating the scores so that the comparison isolates reference-measure effects rather than scale-fitting artifacts. Since the resulting Fisher matrices are nearly isotropic, we summarize them by $\frac{1}{3} \text{tr}(\hat{I})$.

In the small-noise regime $\tau \ll 1$, using the local quadratic expansion of $-\log J_{\text{exp}}(v)$ near the identity, the analytic prediction is

$$I_E = \frac{1}{\tau^2} I_3, \quad I_G = \left(\frac{1}{\tau^2} + \frac{1}{6} \right) I_3, \quad \frac{I_G - I_E}{I_E} = \frac{\tau^2}{6},$$

providing a clean leading-order diagnostic of the Jacobian-induced correction. Table I presents the validation results across concentration parameters spanning two orders of magnitude. Since $\hat{I}_E$ uses $s_E(v) = -v/\tau^2$ while expectations are taken under f_G , one has $\hat{I}_E \approx (1/\tau^4) \mathbb{E}_{f_G}[vv^\top]$, so deviations from $1/\tau^2$ reflect Jacobian-induced non-Gaussian second moments rather than Monte Carlo error. The close agreement with the Laplace prediction for $\tau \lesssim 0.2$ validates the correction as a genuine local geometric effect, while deviations at $\tau \gtrsim 0.5$ indicate higher-order curvature contributions outside the quadratic regime.

Throughout this paper we focus on $SU(2) \cong S^3$, so all concrete charts are three-dimensional; we occasion-

ally write d for dimension-independent statements, but in applications here $d = 3$. Fisher information probes only infinitesimal structure, so it does not by itself determine whether reference-measure mismatch creates global degeneracies or practical optimization failures. We therefore next analyze stationary-point alignment and then quantify the operational consequences via a multi-start optimization study.

Proposition 1 (Stationary-point non-equivalence under reference-measure mismatch). *Let D denote observed data and let*

$$L_E(v) = L_D(v) + R(v), \quad L_D(v) := -\log \mathbb{P}(D | U(v)),$$

be a smooth coordinate-space objective in exponential coordinates $v \in \mathbb{R}^3$. Define the Haar-consistent objective

$$L_G(v) = L_E(v) - \log J_{\text{exp}}(v),$$

where J_{exp} is the chart-volume factor relating Haar measure on $SU(2)$ to Lebesgue measure in v . If v^ satisfies $\nabla L_E(v^*) = 0$ and $\nabla \log J_{\text{exp}}(v^*) \neq 0$, then:*

(i) v^* is not a stationary point of L_G , and

$$\nabla L_G(v^*) = -\nabla \log J_{\text{exp}}(v^*).$$

(ii) The direction $-\nabla L_G(v^*)$ is a strict descent direction for L_G .

(iii) The intrinsic gradient magnitude satisfies

$$\|\nabla L_G(v^*)\| = \|\nabla \log J_{\text{exp}}(v^*)\|,$$

which is generically $O(\|v^\|)$ and need not be small even when $\|\nabla L_E(v^*)\|$ meets optimizer tolerances.*

Proof. By definition,

$$\nabla L_G(v) = \nabla L_E(v) - \nabla \log J_{\text{exp}}(v).$$

If $\nabla L_E(v^*) = 0$, then $\nabla L_G(v^*) = -\nabla \log J_{\text{exp}}(v^*) \neq 0$, proving (i). For (ii), the directional derivative of L_G along $-\nabla L_G(v^*)$ is

$$\left. \frac{d}{d\epsilon} L_G(v^* - \epsilon \nabla L_G(v^*)) \right|_{\epsilon=0} = -\|\nabla L_G(v^*)\|^2 < 0,$$

so $-\nabla L_G(v^*)$ is a strict descent direction for sufficiently small $\epsilon > 0$. Statement (iii) follows immediately from (i). $\square$

For later numerical diagnostics, we use $\|\nabla L_G(v_E^*)\|$ for the intrinsic gradient magnitude evaluated at a flat-coordinate stationary point v_E^* , $d_G(v) := -\nabla L_G(v)/\|\nabla L_G(v)\|$ for the unit intrinsic descent direction, and $\Delta_J(v) := -\nabla \log J_{\text{exp}}(v)$ for the Jacobian correction term omitted by flat-coordinate objectives.

Geometric stationary-point mismatch and surrogate-intrinsic misalignment

The Jacobian J_{chart} is the volume density ratio (Radon–Nikodym derivative) relating Haar measure on $SU(2)$ to Lebesgue measure in a chosen chart. The same object appears as $\sqrt{\det g}$ (metric determinant), as a chart volume factor in change-of-variables formulae.

In exponential coordinates near the identity,

$$L_G(v) = L_E(v) - \log J_{\text{exp}}(v),$$

$$\log J_{\text{exp}}(v) = -\frac{1}{12}\|v\|^2 + O(\|v\|^4),$$

so

$$\nabla L_G(v) = \nabla L_E(v) + \frac{1}{6}v + O(\|v\|^3).$$

If v_E^* satisfies $\nabla L_E(v_E^*) = 0$, then

$$\nabla L_G(v_E^*) = \frac{1}{6}v_E^* + O(\|v_E^*\|^3),$$

which is generically nonzero unless $v_E^* = 0$. Thus, even arbitrarily close to the identity, a stationary point of the flat-coordinate surrogate need not be stationary for the Haar-consistent objective.

A small step taken along the flat-coordinate gradient,

$$\Delta v = -\eta \nabla L_E(v), \quad \eta > 0,$$

decreases L_E to first order. The first-order change in L_G is

$$\begin{aligned} L_G(v + \Delta v) - L_G(v) &= -\eta \nabla L_G(v)^\top \nabla L_E(v) + o(\eta) \\ &= -\eta \left(\|\nabla L_E(v)\|^2 + \frac{1}{6}v^\top \nabla L_E(v) \right) + o(\eta). \end{aligned} \quad (8)$$

Hence, a step that provably decreases L_E can increase L_G whenever the Jacobian-induced term dominates, for example when

$$v^\top \nabla L_E(v) < -6\|\nabla L_E(v)\|^2.$$

This is the practical mechanism behind Proposition 1: reference-measure mismatch introduces a systematic radial component to the gradient field that is absent from the flat surrogate, so optimizer convergence on L_E does not certify stationarity (or even descent) for L_G .

The following section quantifies how often this stationary-point misalignment produces operationally significant errors in a realistic gate-estimation pipeline.

COMPUTATIONAL STUDY: MULTI-START GATE ESTIMATION

We now demonstrate that the gradient-field mismatch identified analytically in Section III produces measurable optimization artifacts across a large-scale multi-start single-qubit gate-estimation study.

We quantify the operational impact of reference-measure mismatch in a realistic single-qubit gate-estimation workflow. Across multi-start L-BFGS-B optimizations of a regularized Born-rule likelihood, flat-coordinate solutions frequently failed to satisfy stationarity for the Haar-consistent objective, even for interior points away from box constraints. These discrepancies produced substantial intrinsic loss reductions and, in some cases, dramatically improved agreement with the ground-truth gate. The effect was strongest in low-data or stress regimes and diminished as the likelihood term dominated.

Although both objectives are defined on $\mathbb{R}^3$, they induce distinct probability measures on $SU(2)$. Consequently, coordinate-space optima of the Euclidean-regularized objective need not coincide with optima of the Haar-consistent formulation.

A. Experimental Design

Single-qubit gates are parametrized in axis-angle (exponential) coordinates $v \in \mathbb{R}^3$ via

$$U(v) = \exp(iv \cdot \sigma/2),$$

with Pauli matrices $\sigma = (\sigma_x, \sigma_y, \sigma_z)$. Writing $v = \|v\|\mathbf{n}$ with $\|\mathbf{n}\| = 1$, this is equivalently $U(v) = \exp(i\|v\|\mathbf{n} \cdot \sigma/2)$. The coordinate vector v is unconstrained; the mapping to physical gates is nonlinear, so $\|v\|$ is not restricted to the Bloch-ball radius.

For a true gate $U_{\text{true}} = U(v_{\text{true}})$, data are generated by sampling random pure input states uniformly on the Bloch sphere, evolving under U_{true} , measuring in the computational basis, and applying independent bit-flip measurement error with probability $p_{\text{flip}} = 0.01$.

This simulation mirrors common calibration and tomography pipelines. Born-rule likelihood evaluation underlies gate-set tomography [3] and randomized benchmarking analyses [5], while Gaussian coordinate-space regularization is widely used as a stabilizing prior in Bayesian gate estimation [6]. Exponential coordinates are standard in control software.

The negative log-likelihood is

$$L_D(v) = -\sum_{\ell=1}^N \log \mathbb{P}(m_\ell \mid \psi_\ell, U(v)), \quad \mathbb{P}(m \mid \psi, U) = |\langle m \mid U \mid \psi \rangle|^2,$$

which is intrinsically defined on $SU(2)$ and therefore coordinate invariant. Reference-measure effects arise solely from regularization, which implicitly specifies a base measure. We compare two MAP objectives,

$$L_E(v) = L_D(v) + \frac{\lambda}{2}\|v\|^2, \quad (9)$$

$$L_G(v) = L_D(v) + \frac{\lambda}{2}\|v\|^2 - \log J_{\text{exp}}(v), \quad (10)$$

where $J_{\text{exp}}(v) = [\sin(\|v\|/2)/(\|v\|/2)]^2$ is the exponential-map Jacobian.

To probe regime dependence, we vary the measurement budget N and regularization strength λ across configurations spanning likelihood-dominated, balanced, and prior-sensitive conditions, as well as stress cases with large parameter norms. Each configuration uses 340 random initializations drawn from $v_0 \sim \mathcal{N}(0, 0.09 I_3)$ and box constraints $v_i \in [-3, 3]$ to maintain chart validity. Additional configurations with randomly sampled interior directions serve as robustness checks against coordinate-specific artifacts. In total, 11,900 converged multi-start runs were analyzed.

B. Ridge-side diagnostics

For each flat-coordinate solution v_E^* satisfying $\|\nabla L_E(v_E^*)\| < \epsilon_{\text{opt}}$ with $\epsilon_{\text{opt}} = 10^{-5}$, we assess intrinsic geometry using complementary diagnostics:

1. **Geometric gradient magnitude:** $\|\nabla L_G(v_E^*)\|$, measuring violation of stationarity under the Haar-consistent objective.
2. **Descent certificate:** For interior points, we take a small step $\Delta v = -\eta \nabla L_G(v_E^*)$ with $\eta = 10^{-3}$ and verify strict decrease $L_G(v_E^* + \Delta v) < L_G(v_E^*)$.
3. **Intrinsic loss gap:** Difference between $L_G(v_E^*)$ and the geometric optimum $L_G(v_G^*)$ obtained from independent optimization.
4. **External accuracy:** Infidelity to the true gate, enabling assessment of practical estimation error.

Solutions on the box boundary are analyzed separately. In such cases, projected gradients may vanish due to constraint saturation even when unconstrained gradients are large; these runs serve as controls for constraint-induced artifacts.

C. Results (empirical overview and quantified extremes)

We evaluate reference-measure mismatch in a multi-start gate-estimation pipeline by comparing Euclidean-regularized optimization (flat objective L_E) with the Haar-consistent objective L_G . In total, 11,900 converged multi-start runs were analyzed. Of these, 11,734 solutions (98.6%) were strictly interior to the box constraints and 166 solutions (1.4%) terminated at the boundary. The principal empirical finding is that *interior* flat-coordinate optima can be strongly non-stationary for the Haar-consistent objective and can lie on ridge-like structures that are absent under L_G . Boundary solutions exhibit a qualitatively different behavior in which $v_E^* = v_G^*$ and intrinsic loss gaps vanish, providing a control that isolates intrinsic mismatch from constraint effects.

Interior ridge-valley mismatch. Interior violations of geometric stationarity were common. Many flat-coordinate optima exhibited substantial $\|\nabla L_G(v_E^*)\|$, positive intrinsic loss gaps, and verified descent directions under L_G , demonstrating that they are not local optima of the intrinsic objective. Table II reports the ten most extreme interior outcomes ranked by $\|\nabla L_G(v_E^*)\|$. These cases show that convergence of the Euclidean-regularized objective does not imply intrinsic stationarity and that the mismatch can be operationally significant, including substantial improvements in physical fidelity after re-optimization under L_G .

The most pronounced improvement occurs in configuration *B1 small*, $\lambda = 10^{-6}$. Here the flat solution has high infidelity despite being strictly interior. This behavior is characteristic of stress configurations: large-norm axis-angle parameters correspond to “long-way-round” representations of rotations that are physically equivalent but geometrically unfavorable under a Haar-consistent objective. The missing Jacobian term introduces a radial bias toward shorter representatives, so geometric re-optimization can move the estimate to a nearby, intrinsically preferred parameterization with much lower infidelity without invoking boundary effects.

Cases where geometric re-optimization yields higher infidelity than the flat solution (negative Diff column in Table II) show differences of at most 0.063, consistent with noise-level variation near a shared basin. In contrast, cases where geometric re-optimization improves fidelity show differences up to 0.848, concentrated in stress configurations with large-norm true parameters. This asymmetry is structurally expected: when the true gate has a compact axis-angle representation, both objectives locate similar solutions and neither is substantially wrong. When the true gate has a large-norm representation, Euclidean regularization introduces a spurious radial bias toward short-norm representatives that is absent from the Haar-consistent objective. The Jacobian term $-\log J_{\text{exp}}(v)$ counteracts this bias, and re-optimization under L_G recovers a geometry-preferred representative with dramatically lower infidelity. The asymmetry in Table II is therefore not a weakness of the geometric objective but a direct signature of the mechanism: the mismatch is consequential precisely where the theory predicts it, and negligible where it does not.

Boundary saturation (constraint-dominated regime). Boundary-terminated runs exhibit a distinct behavior. Although the unconstrained intrinsic gradient $\|\nabla L_G(v_E^*)\|$ can be extremely large at these points, Euclidean and Haar-consistent optimizations converge to the same feasible solution ($v_E^* = v_G^*$), yielding zero intrinsic loss gap. This outcome reflects box saturation: steps that would reduce the intrinsic objective lie outside the feasible region and are projected back to the same boundary point. The Jacobian correction therefore alters the local gradient but cannot produce a different estimate. These runs confirm that the large discrepancies observed in interior cases are not artifacts of the box

TABLE II. Top 10 interior extremes ranked by $\|\nabla L_G(v_E^*)\|$. All cases are strictly interior (no box constraints active). Configuration labels use the form “Letter-Index Regime, λ ,” e.g., *B1 small*, $\lambda = 10^{-6}$. B denotes boundary-stress truth configurations (large-norm parameters), C denotes baseline interior truth configurations, and R denotes randomized interior controls. Regime tags indicate data conditions: *small* = moderate data with stress-level truth; *lo* = low-data regime; *hi* = high-data regime. Columns report intrinsic gradient magnitude, gate infidelity under flat and geometric solutions, intrinsic loss gap, parameter displacement between solutions, and distances from the true parameter. Negative values in the Diff column (flat – geometric) indicate cases where geometric re-optimization yields marginally higher infidelity; these differences are at most 0.063 and reflect noise-level variation near a shared solution basin. Positive values, concentrated in stress configurations, reflect the directional bias introduced by Euclidean regularization in large-norm regimes.

Config	$\ \nabla L_G\ $	Flat Infid	Geom Infid	Diff	Gap	$\ v_E - v_G\ $	Dist _E	Dist _G
B1 small, $\lambda = 10^{-6}$	34.633	0.872	0.024	0.848	24.434	3.524	3.220	0.420
C2, $\lambda = 10^{-6}$	1.280	0.049	0.112	-0.063	2.001	5.643	6.055	0.743
C2, $\lambda = 0$	1.280	0.049	0.112	-0.063	2.001	5.643	6.055	0.743
C2 lo, $\lambda = 10^{-6}$	1.126	0.075	0.131	-0.056	1.658	5.672	5.840	0.823
C2 lo, $\lambda = 10^{-6}$	1.112	0.087	0.131	-0.044	1.632	5.626	5.809	0.823
C2, $\lambda = 10^{-6}$	1.007	0.300	0.112	0.188	1.504	4.896	5.448	0.743
C2, $\lambda = 0$	1.007	0.300	0.112	0.188	1.504	4.896	5.448	0.743
C2 lo, $\lambda = 10^{-6}$	0.926	0.931	0.131	0.799	1.271	4.943	4.208	0.823
B3 small, $\lambda = 10^{-6}$	0.925	0.997	0.900	0.097	0.153	1.045	4.540	3.706
B3 small, $\lambda = 10^{-6}$	0.920	0.995	0.900	0.095	0.143	0.969	4.485	3.706

constraint.

Configuration-level structure. Table III summarizes regime dependence across configurations using boundary incidence and maximum gradient magnitudes. Stress and low-data configurations produce the largest interior violations, whereas high-data settings exhibit small interior gradients with boundary events either absent or confined to a small subset of starts. Random-direction interior controls show the same qualitative pattern: occasional boundary saturation with large unconstrained gradients but modest interior mismatch otherwise. These statistics indicate that the mismatch is structural and regime dependent, becoming most pronounced when regularization meaningfully shapes the objective relative to the likelihood.

Summary The computational study supports three conclusions. First, flat-coordinate convergence can yield interior solutions that are strongly non-stationary for the Haar-consistent objective, producing large intrinsic loss gaps and, in extreme cases, substantial improvements in physical fidelity after geometric re-optimization. Second, boundary terminations constitute a distinct constraint-dominated regime in which $v_E^* = v_G^*$ and loss gaps vanish despite very large unconstrained gradients, confirming that the strongest operational effects arise in the interior. Third, the severity of mismatch is regime dependent and is amplified in stress or low-data settings where regularization competes with likelihood curvature.

Magnitude and practical scope of the effect

Locally, the Jacobian induces a relative Fisher-information correction of order $\tau^2/6$, which is negligible for highly concentrated posteriors but becomes non-

trivial once uncertainty approaches the curvature scale of $SU(2)$. Globally, the multi-start study shows that the same mechanism is structurally present even when the practical fidelity impact is modest: in all interior runs, flat-coordinate convergences were non-stationary for the Haar-consistent objective and exhibited clear intrinsic descent directions toward lower intrinsic loss. In regimes with strong regularization or proximity to the chart boundary, the effect is no longer perturbative and can produce very large gradients and substantial deviations from the intrinsic optimum, driven by constraint-dominated behavior near the singularity of the exponential map.

Coordinate-change principle for gate models

A gate model is intrinsically a probability model on the group $SU(2)$ (or $SU(d)$), while any parametrization $v \in \mathbb{R}^3$ is merely a chart used to compute with that model. If $U : \mathbb{R}^3 \supset \mathcal{V} \rightarrow SU(2)$ is a smooth local parametrization (e.g., exponential coordinates), then the physically meaningful likelihood for data D is a function of the group element,

$$\mathbb{P}(D | v) = \mathbb{P}(D | U(v)),$$

and is therefore invariant under reparametrization of v (the Born-rule likelihood and related intrinsic data terms used in tomography and benchmarking are of this form [5]).

The subtlety is that a *regularized* or *MAP* objective is not determined by the likelihood alone: it also encodes a choice of reference measure (or base measure) for the parameter prior. In particular, the common coordinate-space quadratic regularizer $\|v\|^2$ corresponds to a density

TABLE III. Per-configuration summary of mismatch structure. Each configuration used 340 multi-start runs. Columns report the number of boundary-terminated runs and the maximum intrinsic gradient magnitude observed among interior and boundary solutions.

Config	Boundary Runs	Max $\ \nabla L_G\ $ (Interior)	Max $\ \nabla L_G\ $ (Boundary)
B1 small	3	34.633	0.792
B2 small	14	0.809	17.095
B3 small	18	0.925	13.330
C1 lo, $\lambda = 10^{-6}$	8	0.427	22.454
C2 lo, $\lambda = 10^{-6}$	6	1.126	3.640
C2, $\lambda = 0$	3	1.280	114.882
C3 hi	3	0.225	461.229
C4 hi	7	0.187	654.953
C5 hi	0	0.330	—

defined relative to Lebesgue measure in the chosen chart, whereas a principled “uninformative” prior on the group is naturally defined relative to Haar measure. These two choices agree only to lowest order near the identity and diverge as curvature becomes non-negligible.

Formally, let μ_H denote Haar measure on $SU(2)$ and let λ denote Lebesgue measure on $\mathbb{R}^3$. In a chart $U(v)$, the pullback of μ_H has the form

$$U^* \mu_H = J_{\text{chart}}(v) d\lambda(v),$$

where $J_{\text{chart}}(v)$ is the chart-volume factor (equivalently $\sqrt{\det g(v)}$ for the induced metric). For exponential coordinates on $SU(2)$ [4], Equation (2) applies.

Consequently, if the intended prior is defined intrinsically as a density $\pi(U) d\mu_H(U)$ on the group, then its coordinate representation is not $\pi(U(v)) dv$ but rather

$$\pi(U(v)) J_{\text{chart}}(v) dv.$$

This yields the coordinate-change principle used throughout the paper:

Coordinate-change principle. *Likelihood terms that are defined on $SU(2)$ are chart-invariant, but any term interpreted as a prior or regularizer must transform with the chart volume factor. Under a change of coordinates, the correct coordinate-space objective gains the additive correction $-\log J_{\text{chart}}(v)$.*

Applied to regularized gate estimation, where the data term is intrinsic and the regularizer is written in coordinates, the two objectives are

$$L_E(v) = L_D(v) + \frac{\lambda}{2} \|v\|^2, \quad (11)$$

$$L_G(v) = L_D(v) + \frac{\lambda}{2} \|v\|^2 - \log J_{\text{exp}}(v), \quad (12)$$

where $L_D(v) = -\log \mathbb{P}(D | U(v))$ is coordinate-invariant (up to numerical discretization) and the $-\log J_{\text{exp}}$ term is the required reference-measure correction when the intended base measure on the group is Haar.

Two clarifications prevent common confusions. First, L_E and L_G are both functions on the *same* coordinate-space $\mathbb{R}^3$; the distinction is not “different coordinates,” but “different induced measures on $SU(2)$.” Second, this is not a cosmetic curvature rescaling: since

$$\nabla L_G(v) = \nabla L_E(v) - \nabla \log J_{\text{exp}}(v),$$

the correction modifies the gradient field and can therefore shift stationary points (Proposition 1), which is the mechanism underlying the discrepancies observed in the multi-start study (Section IV).

Finally, the connection to standard practice is straightforward. Tomography and randomized benchmarking employ intrinsically defined likelihoods [5]. At the same time, Bayesian and regularized estimators frequently impose coordinate-space priors and quadratic penalties in Lie-algebra coordinates for computational convenience [6]. The coordinate-change principle above pinpoints when that convenience becomes a modeling assumption (Lebesgue base measure in the chart), and when the Haar-consistent alternative introduces a non-negligible correction through $J_{\text{exp}}(v)$.

V. IMPLICATIONS FOR INFERENCE, OPTIMIZATION, AND FISHER INFORMATION

Geometric distortion does not merely bias curvature estimates; it alters the objective itself whenever a surrogate loss is written in flat coordinates while the intended statistical model is defined with respect to an intrinsic volume form. In such cases the coordinate-space objective is not a reparametrization of the same surface but a different surface on $SU(2)$. Including the chart-volume density restores the intended reference measure and therefore the correct intrinsic geometry of the likelihood. Omitting it modifies the gradient field, shifts stationary points, and can change convergence behavior. Although optimization is performed in coordinate space, optimality is defined on the group; differences in convergence therefore reflect differences in the induced objective on $SU(2)$ rather than artifacts of parametrization.

Riemannian versus Euclidean Fisher information

Let (M, g) be a Riemannian manifold (here $M = SU(2)$ or S^2) equipped with Haar measure μ_{Haar} . For a one-parameter family of probability densities $p(\omega \mid \vartheta)$ defined intrinsically on M , the coordinate-invariant Fisher information is [7]

$$I_R(\vartheta) = \int_M (\partial_\vartheta \log p(\omega \mid \vartheta))^2 p(\omega \mid \vartheta) d\mu_{\text{Haar}}(\omega). \quad (13)$$

If $y = \phi(\omega)$ denotes a local coordinate chart, then the Haar measure pulls back as $d\mu_{\text{Haar}}(\omega) = J_\phi(y) dy$, and the induced coordinate density with respect to Lebesgue measure is

$$f_Y(y \mid \vartheta) = p(\phi^{-1}(y) \mid \vartheta) J_\phi(y).$$

When this induced density is used, Fisher information computed in coordinates agrees with Eq. (13), reflecting true coordinate invariance within a fixed statistical model.

In contrast, many practical surrogates replace $f_Y(y \mid \vartheta)$ by a simpler coordinate-space density that omits the Jacobian factor and implicitly treats Lebesgue measure as the reference measure. This is not a coordinate reparametrization of the same model but a change of statistical model through a different choice of reference measure, as confirmed by the empirical mismatch between flat-coordinate and Haar-consistent optima. As demonstrated by the Lie-algebra example, discarding the Jacobian alters not only Fisher information but also the gradient field and stationary-point structure of the objective, producing extrema that do not correspond to extrema of the intrinsic Haar-consistent likelihood.

Model misspecification versus coordinate choice

This distinction is particularly consequential in Bayesian and MAP-based pipelines. Even when the data likelihood $\mathbb{P}(D \mid U)$ is defined intrinsically and is therefore coordinate invariant—as is the case for Born-rule likelihoods—the posterior is not invariant once a prior density is specified. A prior that is flat in local coordinates implicitly adopts Lebesgue measure as its reference geometry and is not Haar-isotropic. Consequently, MAP estimation with a coordinate-flat prior optimizes a different objective than MAP estimation with a Haar-consistent prior, even when the likelihood itself is intrinsic. The computational study demonstrates that this difference can produce interior solutions that are stationary for the coordinate objective but not for the intrinsic one.

The discrepancy quantified in Table I reflects this distinction: f_E and f_G are distinct statistical models, not different coordinate representations of the same one. They assign probability to the same group elements using different reference measures. The difference appears

already at second order in the log-density and persists arbitrarily close to the identity.

It is therefore essential to distinguish two geometric choices that are often conflated in parameter-space learning. The first is the choice of reference measure, which determines the statistical model itself (what is meant by isotropy, likelihood, and prior). The second is the choice of metric used by the optimizer to convert gradients into steps. The present paper addresses the former. Even when optimization is performed using Euclidean methods, correcting the reference measure via the Jacobian restores the intended intrinsic objective. Natural-gradient methods address the latter, but they operate on whatever statistical model has been specified.

Within a single statistical model, Fisher information is invariant under smooth reparametrization. Here, however, the objective itself changes: the Jacobian contributes terms that shift gradients and Hessians, so that coordinate-space ascent can correspond to intrinsic descent. In curved parameter spaces, Euclidean surrogates can therefore converge to points that are not merely biased but qualitatively incorrect with respect to the intended intrinsic likelihood, even when the solution lies well within the coordinate chart.

Practical optimizer fix: Jacobian-corrected gradients in Euclidean optimizers

Because optimization objectives are typically written as losses to be minimized rather than log-densities to be maximized, the Jacobian enters with a minus sign in the geometry-consistent loss. If a surrogate objective is written in local coordinates $y \in \mathbb{R}^d$ (here y denotes generic chart coordinates, not specifically stereographic) as $L_E(y)$ while the intended statistical model is defined with respect to an intrinsic reference measure, the reference-measure-consistent objective is

$$L_G(y) = L_E(y) - \log J_{\text{chart}}(y), \quad (14)$$

where $J_{\text{chart}}(y)$ denotes the chart-volume factor.

Accordingly, the primary modification required in a Euclidean optimizer is to replace the surrogate gradient by the Jacobian-corrected gradient,

$$\nabla L_G(y) = \nabla L_E(y) - \nabla \log J_{\text{chart}}(y). \quad (15)$$

This correction is model-based rather than algorithm-specific and therefore applies equally to SGD, Adam, RMSProp, and quasi-Newton methods.

Reference-Measure Selection in Parameter-Space Optimization

The chart-volume factor $J(v)$ or $\sqrt{\det g(y)}$ should be included when:

- Perturbations, priors, or regularizers are intended to be isotropic with respect to intrinsic (Haar/Riemannian) geometry
- The objective is a surrogate loss, regularized MAP estimate, or penalized likelihood formulated in local coordinates
- Optimization convergence behavior and stationary-point locations are relevant to the inference task

Omitting the chart-volume factor specifies a distinct statistical model:

- The flat-coordinate objective corresponds to a different gradient field and distinct stationary points
- The difference is not a coordinate reparametrization but a change of reference measure on parameter space
- Ascent directions in coordinate-space can correspond to descent directions under the intrinsic objective
- Local extrema and saddle points generically differ between flat-coordinate and geometry-consistent models

The chart-volume factor may be omitted when:

- Physical noise or control errors are genuinely coordinate-Gaussian
- Flat-coordinate geometry is explicitly intended to define isotropy
- The statistical model is meant to be Lebesgue-based rather than Haar-based

Practical Optimizer Fix

Step 1. Choose the intended reference measure (Haar/Riemannian vs. chart Lebesgue).

Step 2. Compute the chart-volume factor $J(y)$.

Step 3. Add $-\log J(y)$ to the loss.

Step 4. Replace ∇L_E with $\nabla L_E - \nabla \log J$.

Optional Step 5. Apply natural-gradient methods *after* correcting the objective.

Unknown or implicit geometry

If the intrinsic measure is not available in closed form, the chart-volume factor must be approximated rather than omitted. One principled approach is to use a pullback Riemannian metric induced by the model map $F : \mathbb{R}^d \rightarrow M$, yielding $J_{\text{chart}}(y) = \sqrt{\det g(y)}$. The corresponding $\nabla \log J(y)$ can be computed analytically when the Jacobian of F is available or via automatic differentiation.

The essential point is that the Jacobian factor is not an optional normalization: it is part of the statistical model whenever isotropy or noninformativeness is defined relative to intrinsic volume. Including $-\log J_{\text{chart}}(y)$ restores the intended reference measure and therefore the correct gradient field; omitting it solves a different optimization problem, regardless of whether the intrinsic optimum lies in the interior or on a constraint boundary.

VI. DISCUSSION

Geometry versus physical noise

The effects identified here arise from replacing the intrinsic reference measure on $SU(2)$ with a flat coordinate-space surrogate. Flat coordinate-space priors are not intrinsically incorrect. When the scientific intent is to model additive noise in control parameters—for example, uncorrelated Gaussian perturbations introduced by hardware imperfections in axis-angle coordinates—a coordinate-Gaussian prior is physically appropriate, and the Euclidean model f_E correctly represents the intended noise structure.

Our analysis applies specifically to workflows that combine intrinsically defined likelihoods (e.g., Born-rule models on $SU(2)$) with coordinate-space regularization while intending isotropy with respect to the group geometry. In such cases, omitting the chart-volume factor does not merely rescale curvature but selects a statistically distinct model with a different reference measure.

In small-noise, high-data regimes, Euclidean surrogates may nevertheless provide accurate local approximations near the identity. The significance of the reference-measure mismatch is therefore structural rather than pointwise: the Jacobian term modifies the objective itself, including its gradient field and stationary-point structure. As a result, stationary points of flat-coordinate surrogates can be non-stationary under the Haar-consistent objective, and coordinate-gradient ascent directions can, in some regimes, correspond to local descent directions under the intrinsic geometry. The computational study shows that such discrepancies can arise for strictly interior solutions with moderate parameter norms, demonstrating that the effect is not confined to chart boundaries or extreme parametrizations.

Bias versus gradient corruption

The $1/6$ coefficient appearing in the second-order expansion of $-\log J_{\text{exp}}(v)$ near the identity is sometimes interpreted as a small numerical correction to objective values. This interpretation understates the impact of reference-measure mismatch. While the coefficient quantifies local curvature bias, the more consequential effect occurs in the gradient field: omitting $-\log J_{\text{exp}}(v)$ removes a term whose gradient $-\nabla \log J_{\text{exp}}(v)$ is generally nonzero away from the identity, radial, and direction dependent.

Accordingly, the stationary-point conditions for the flat-coordinate surrogate,

$$\nabla[-\log \mathbb{P}(D | U(v)) + \frac{\lambda}{2}\|v\|^2] = 0,$$

and for the geometry-consistent objective,

$$\nabla[-\log \mathbb{P}(D | U(v)) + \frac{\lambda}{2}\|v\|^2 - \log J_{\text{exp}}(v)] = 0,$$

are not equivalent even arbitrarily close to the identity. Because the Jacobian term contributes directly to the gradient, its omission alters the vector field governing optimization. Flat-coordinate surrogates can therefore converge to parameter values that satisfy the flat optimality conditions yet remain far from stationary for the intrinsic objective, with $\|\nabla L_G(v_E^*)\|$ remaining substantial relative to convergence tolerances.

In this sense, the $1/6$ coefficient measures only local curvature bias in objective values, whereas the missing gradient term changes the direction field itself.

Why this is not just a natural-gradient result

Natural gradient methods respect the information geometry of the statistical model that has been specified. Consequently, if that model is defined using a coordinate-space reference measure—such as a Gaussian prior in parameters that omits the chart-volume factor—the same reference-measure mismatch identified here is inherited by the natural gradient. The choice of reference measure (which model is specified) is therefore logically prior to the choice of optimizer metric (how it is optimized).

The present work addresses this prior issue: specification of the objective itself. Natural gradient methods precondition updates using the Fisher information metric, but they operate on whatever loss function has been defined. If that loss encodes a flat-coordinate prior when Haar isotropy was intended, natural-gradient descent optimizes the coordinate-space model rather than the intrinsic one.

For regularized MAP estimation,

$$L_E(v) = L_D(v) + \frac{\lambda}{2}\|v\|^2,$$

a natural-gradient optimizer uses the Fisher metric associated with this objective, implicitly treating the

quadratic penalty as part of the model. If isotropy with respect to Haar measure is intended, the appropriate objective is

$$L_G(v) = L_D(v) + \frac{\lambda}{2}\|v\|^2 - \log J_{\text{exp}}(v).$$

Natural gradient applied to L_E therefore optimizes a different statistical model than either Euclidean or natural gradient applied to L_G . The reference measure determines the model; the optimizer determines how that model is traversed.

Where this arises in practice

The mismatch identified here does not arise when inference relies solely on intrinsically defined likelihoods, such as standard Born-rule tomography. It becomes relevant whenever priors, regularizers, or surrogate objectives are formulated directly in parameter space.

Such situations occur in many workflows on curved parameter manifolds [9, 10], including regularized gate-set tomography, Bayesian gate estimation with Lie-algebra or Euler-angle priors, and learning-based control objectives that penalize deviations in axis-angle coordinates using Euclidean norms. In these settings, “isotropic noise” or “uninformative priors” implemented in coordinates implicitly select a flat chart reference measure on $SU(2)$, making the reference measure part of the statistical model rather than a mere parametrization choice.

Concrete example: GST with Lie-algebra priors

In gate-set tomography workflows, Gaussian priors are often imposed on Lie-algebra parameters $v \in \mathbb{R}^3$, yielding

$$L_E(v) = -\log \mathbb{P}(D | U(v)) + \frac{\lambda}{2}\|v\|^2.$$

If isotropy with respect to Haar measure is intended, the geometry-consistent objective is

$$L_G(v) = -\log \mathbb{P}(D | U(v)) + \frac{\lambda}{2}\|v\|^2 - \log J_{\text{exp}}(v).$$

For calibration uncertainties $\tau \sim 0.1\text{--}0.5$ rad, omitting $\log J_{\text{exp}}(v)$ modifies posterior curvature and can shift stationary-point locations, with effects typically modest near the identity but increasing as uncertainty approaches the curvature scale of the group.

Limitations

This work is restricted to single-qubit systems. Extending the analysis to $SU(2^n)$ would require explicit treatment of the intrinsic metric and volume form, which depend on entanglement structure. Our validation also

focuses on axis–angle parametrization and quadratic regularization; alternative coordinate systems or priors may change the magnitude but not the mechanism of the mismatch.

Scope and generalization

Although derived for single-qubit geometry, the underlying mechanism is general. Whenever inference or optimization is performed in local coordinates on a curved parameter manifold, omitting the chart-volume factor changes the reference measure and therefore the objective being optimized.

For independently parametrized qubits, the correction factorizes,

$$J_{\text{exp}}(v_1, \dots, v_n) = \prod_{j=1}^n J_{\text{exp}}(v_j),$$

and scales linearly with system size. Global gates and entangled states, however, do not admit such factorization and require separate analysis.

The central message is therefore general: when isotropy or noninformativeness is defined relative to intrinsic geometry, implementing that intent using flat coordinate-space surrogates changes the statistical model. In such settings, coordinate-space optima can remain non-stationary under the intrinsic objective, reflecting the mismatch between the two gradient fields.

DISCLOSURES

Competing Interests. The authors have no relevant financial or non-financial interests to disclose.

Code and Results. All code and results will be made publicly available upon acceptance.

Author Contributions. All authors contributed equally to the conceptualization, methodology, software development, analysis, visualization, and writing of this manuscript.

Use of Artificial Intelligence Tools. Generative artificial intelligence tools were used to assist with prose refinement, L^AT_EX formatting, and figure layout. All analytical decisions, code implementation, validation, and interpretation of results were performed by the authors, who take full responsibility for the accuracy, originality, and integrity of the work.

Ethics Approval and Consent to Participate. This study does not involve human participants, human data, or animal subjects and therefore did not require ethical approval or informed consent.

Appendix A: Stereographic Projection and Induced Coordinate Densities

This appendix gives a closed-form *global* illustration of reference-measure distortion under stereographic projection on $S^3 \cong SU(2)$, intended purely as an intuition aid to make the chart-volume mechanism visually explicit (Fig. 2). Stereographic coordinates are not used in practical quantum workflows; all quantitative predictions and numerical experiments in the main text rely exclusively on the exponential (axis–angle) chart, which is the operative coordinate system for single-qubit control and tomography.

1. Stereographic projection on S^3

The group $SU(2)$ is diffeomorphic to the 3-sphere $S^3 \subset \mathbb{R}^4$ [11] via

$$U = \begin{pmatrix} \alpha & -\bar{\beta} \\ \beta & \bar{\alpha} \end{pmatrix}, \quad |\alpha|^2 + |\beta|^2 = 1. \quad (\text{A1})$$

Stereographic projection $\pi : S^3 \setminus \{N\} \rightarrow \mathbb{R}^3$ from the north pole $N = (0, 0, 0, 1)$ is

$$\pi(\omega) = \frac{(\omega_1, \omega_2, \omega_3)}{1 - \omega_4}, \quad \omega \in S^3. \quad (\text{A2})$$

Solving for ω yields the inverse map [12]. Writing $y = \pi(\omega)$ and $r^2 = \|y\|^2$,

$$\pi^{-1}(y) = \frac{1}{1 + r^2} (2y, r^2 - 1). \quad (\text{A3})$$

Jacobian density and volume distortion

Because $\pi^{-1} : \mathbb{R}^3 \rightarrow S^3 \subset \mathbb{R}^4$ maps into a 3-dimensional submanifold, the appropriate change-of-variables factor is given by the Gram determinant (equivalently, the square root of the determinant of the pulled-back metric). Define

$$J_{\pi}(y) = \sqrt{\det g(y)} = \sqrt{\det((D\pi^{-1}(y))^{\top} D\pi^{-1}(y))}. \quad (\text{A4})$$

Stereographic projection is conformal with conformal factor $\lambda(y) = 2/(1 + \|y\|^2)$ [12, Prop. 1.26], so the pullback metric satisfies $g(y) = \lambda(y)^2 I_3$ and hence $J_{\pi}(y) = \lambda(y)^3$, giving

$$d\mu_g = J_{\pi}(y) dy = \left(\frac{2}{1 + \|y\|^2} \right)^3 dy. \quad (\text{A5})$$

Induced coordinate densities

Proposition 2 (Induced density under stereographic projection). *Let X be distributed on S^3 with density*

$f_{\text{Haar}}(\omega) \in C^\infty(S^3)$, $f_{\text{Haar}} > 0$, with respect to Haar measure μ_{Haar} on S^3 , and let $Y = \pi(X)$. Then the induced density of Y with respect to Lebesgue measure μ_{Leb} on $\mathbb{R}^3$ is

$$f_{Y,\text{Leb}}(y) = f_{\text{Haar}}(\pi^{-1}(y)) \left(\frac{2}{1 + \|y\|^2} \right)^3, \quad y \in \mathbb{R}^3. \quad (\text{A6})$$

If $f_{\text{Haar}}(N) > 0$, then as $\|y\| \rightarrow \infty$,

$$f_{Y,\text{Leb}}(y) \sim 8 f_{\text{Haar}}(N) \|y\|^{-6}.$$

For the Haar-uniform case $f_{\text{Haar}}(\omega) \equiv 1/\text{Vol}(S^3)$ (with $\text{Vol}(S^3) = 2\pi^2$), the induced Lebesgue density has tail constant $8/\text{Vol}(S^3) = 4/\pi^2$.

Proof. For any measurable $A \subset \mathbb{R}^3$,

$$\mathbb{P}(Y \in A) = \mathbb{P}(X \in \pi^{-1}(A)) = \int_{\pi^{-1}(A)} f_{\text{Haar}}(\omega) d\mu_g(\omega).$$

Pulling back via π^{-1} and using (A5) yields

$$\mathbb{P}(Y \in A) = \int_A f_{\text{Haar}}(\pi^{-1}(y)) \left(\frac{2}{1 + \|y\|^2} \right)^3 dy,$$

establishing (A6). As $\|y\| \rightarrow \infty$, one has $\pi^{-1}(y) \rightarrow N$, and smoothness gives $f_{\text{Haar}}(\pi^{-1}(y)) \rightarrow f_{\text{Haar}}(N) > 0$, which implies the stated tail behavior. $\square$

Interpretation

For Haar-uniform f_{Haar} , the induced coordinate density satisfies

$$f_{Y,\text{Leb}}(y) \propto \|y\|^{-6} \quad \text{as } \|y\| \rightarrow \infty.$$

This heavy-tailed appearance is *entirely* a coordinate artifact: the intrinsic distribution on S^3 remains uniform with respect to Haar volume. Stereographic projection therefore provides a global, closed-form illustration of the same reference-measure mechanism that appears locally in exponential coordinates, where it manifests as the Jacobian factor $J_{\text{exp}}(v)$ used throughout the main text.

Appendix B: Reference-Measure Pullback and Induced Coordinate Densities

This appendix records the standard measure-theoretic relation underlying the main text: an intrinsic probability density defined with respect to Riemannian (Haar) volume acquires a chart-volume factor when expressed as a Lebesgue density in local coordinates. Omitting this factor therefore changes the reference measure and defines a different statistical model.

Let (M, g) be an n -dimensional Riemannian manifold with Riemannian volume measure μ_g , and let $\psi : V \subset \mathbb{R}^n \rightarrow M$ be a smooth coordinate map. The pullback of the volume measure satisfies

$$\psi^*(d\mu_g) = \sqrt{\det g(y)} dy, \quad (\text{B1})$$

where $g(y)$ is the metric tensor expressed in coordinates and dy is Lebesgue measure on $\mathbb{R}^n$.

Proposition 3 (Induced coordinate density). *If a random variable X on M has density $p_X(\omega)$ with respect to μ_g , and $Y = \psi^{-1}(X)$ denotes its coordinate representation, then Y has Lebesgue density*

$$f_Y(y) = p_X(\psi(y)) \sqrt{\det g(y)}. \quad (\text{B2})$$

Proof. For measurable $B \subset V$,

$$\mathbb{P}(Y \in B) = \int_{\psi(B)} p_X(\omega) d\mu_g(\omega) = \int_B p_X(\psi(y)) \sqrt{\det g(y)} dy,$$

using (B1). $\square$ $\square$

For a compact Lie group endowed with a bi-invariant Riemannian metric (e.g., $SU(2)$), the associated Riemannian volume form is left- and right-invariant and therefore coincides, up to normalization, with Haar measure; see, e.g., [13, Ch. I, Secs. 2–3], [4, Ch. 4], and for the construction of the Riemannian volume form, [12, Chs. 15–16].

Proposition 3 therefore applies directly to intrinsic group-valued probability models. Omitting the chart-volume factor in coordinates replaces Haar volume by Lebesgue measure and hence specifies a different statistical model, not a reparametrization of the same one.

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- [1] J. J. Sakurai and Jim Napolitano. *Modern Quantum Mechanics*. Cambridge University Press, 2 edition, 2017.
 - [2] Daniel F. V. James, Paul G. Kwiat, William J. Munro, and Andrew G. White. Measurement of qubits. *Physical Review A*, 64(5):052312, 2001. Maximum-likelihood estimation for quantum state tomography using Euclidean parameterization.
 - [3] Erik Nielsen, Robin Blume-Kohout, John K. Gamble, and Kenneth M. Rudinger. Gate set tomography. *Quantum Science and Technology*, 6(4):044006, 2021.
 - [4] Brian C. Hall. *Lie Groups, Lie Algebras, and Representations*. Springer, 2 edition, 2015. Compact Lie groups, Haar measure, and $SU(2)$.
 - [5] Easwar Magesan, Jay M. Gambetta, and Joseph Emerson. Efficient measurement of quantum gate error by interleaved randomized benchmarking. *Physical Review Letters*, 109(8):080505, 2012. Gate calibration via exponential curve fitting to decay rates using Euclidean loss.
 - [6] Ian Hincks, Joel J. Wallman, Chris Ferrie, Chris Granade, and David G. Cory. Bayesian inference for

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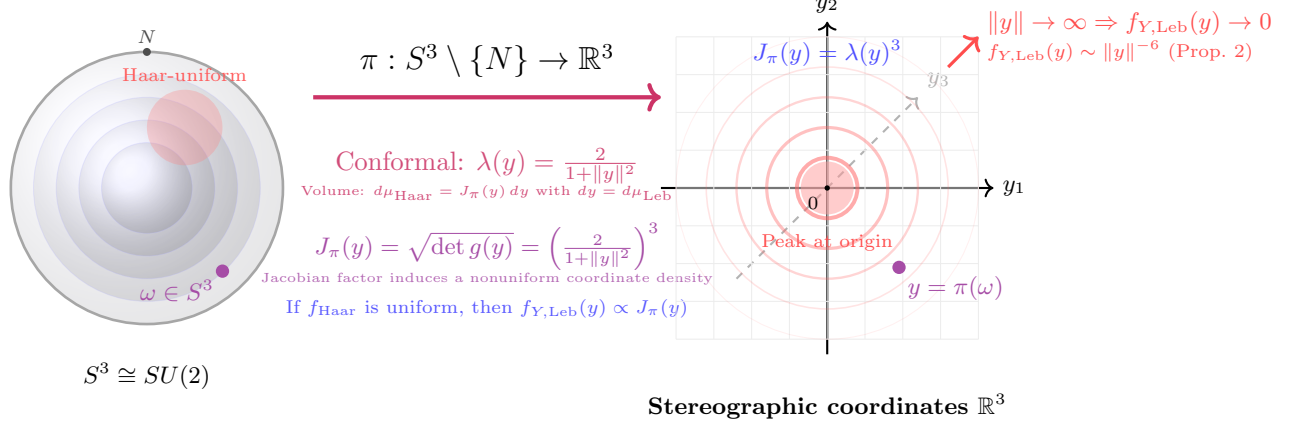


FIG. 2. Stereographic projection $\pi : S^3 \setminus \{N\} \rightarrow \mathbb{R}^3$ and the induced reference-measure distortion. A Haar-uniform intrinsic distribution becomes nonuniform in coordinates due solely to the chart-volume factor $J_\pi(y) = \left(\frac{2}{1+\|y\|^2}\right)^3$.

randomized benchmarking protocols. *arXiv preprint arXiv:1802.00401*, 2018.

- [7] Shun-ichi Amari and Hiroshi Nagaoka. *Methods of Information Geometry*, volume 191 of *Translations of Mathematical Monographs*. American Mathematical Society, Providence, RI, 2000.
- [8] Shun-Ichi Amari. Natural gradient works efficiently in learning. *Neural computation*, 10(2):251–276, 1998.
- [9] Mark Girolami and Ben Calderhead. Riemann manifold langevin and hamiltonian monte carlo methods. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, 73(2):123–214, 2011.
- [10] Simon Byrne and Mark Girolami. Geodesic monte carlo

on embedded manifolds. *Scandinavian Journal of Statistics*, 40(4):825–845, 2013.

- [11] Gregory L. Naber. *Topology, Geometry and Gauge Fields: Foundations*. Springer, 2 edition, 2011. Diffeomorphisms between $SU(2)$ and the 3-sphere.
- [12] John M. Lee. *Introduction to Smooth Manifolds*, volume 218 of *Graduate Texts in Mathematics*. Springer, New York, 2 edition, 2013. Stereographic projection and volume forms on S^n .
- [13] Sigurdur Helgason. *Differential Geometry, Lie Groups, and Symmetric Spaces*, volume 34 of *Graduate Studies in Mathematics*. American Mathematical Society, Providence, RI, 2001.